



CS & IT ENGINEERING

Discrete Mathematics

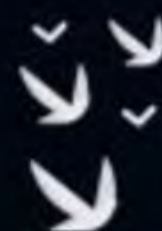
Graph Theory

Lecture No.- 14



By- Satish Sir

Recap of Previous Lecture



Topic

Matching Set Part-01



Topics to be Covered



Topic

Matching Set Part-02

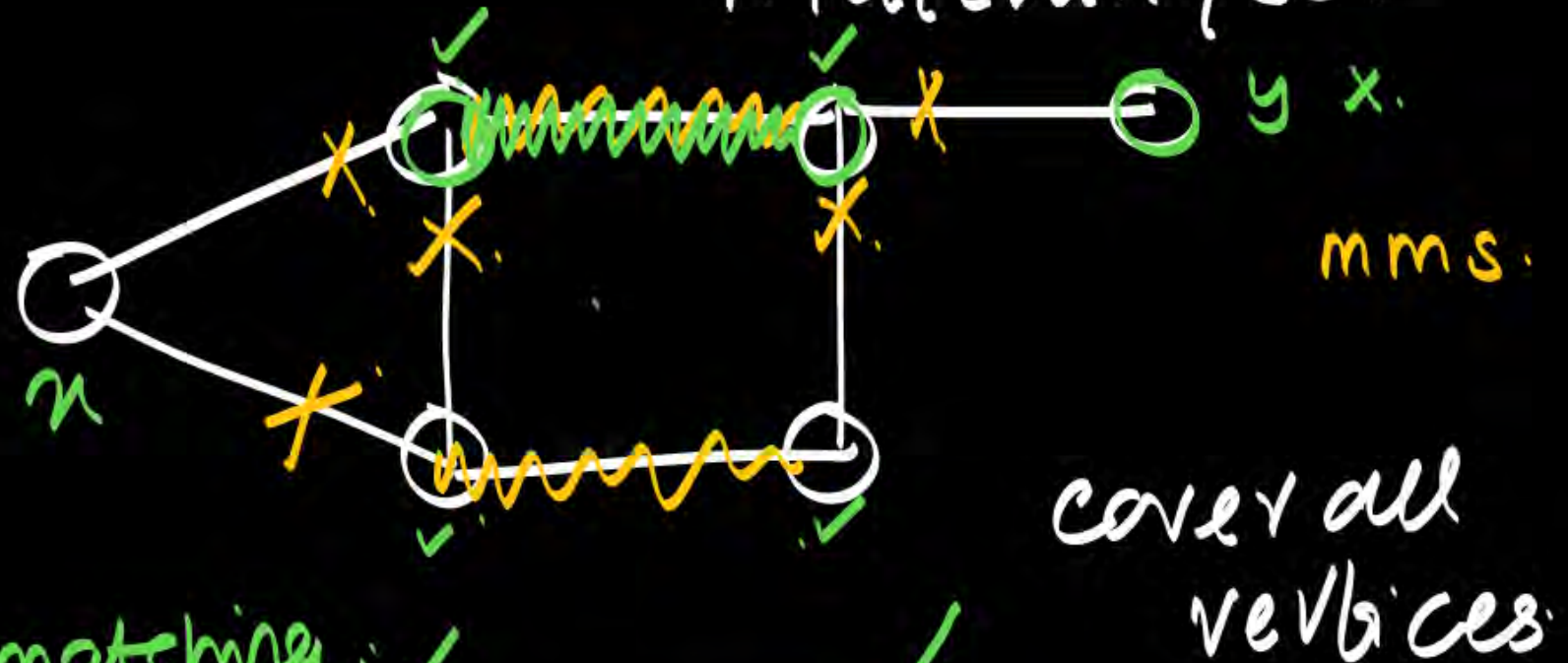


perfect matching:

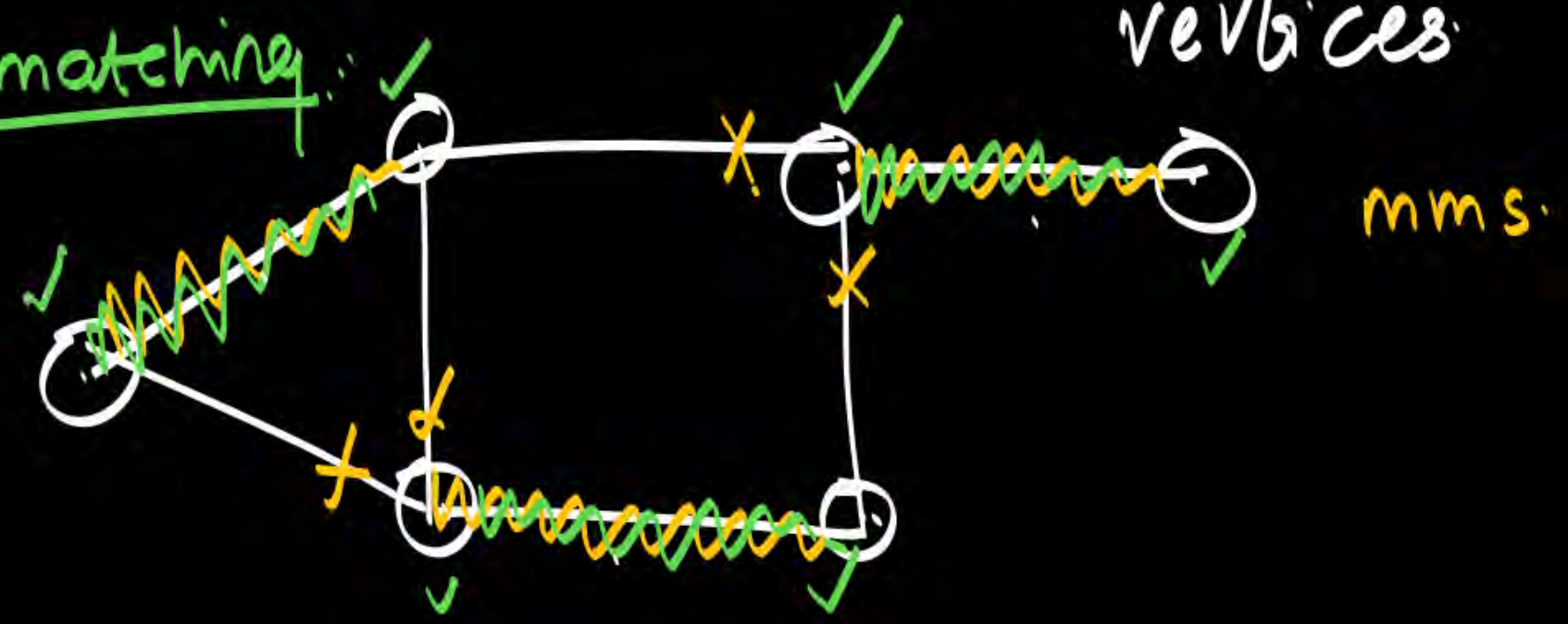
maximal matching set

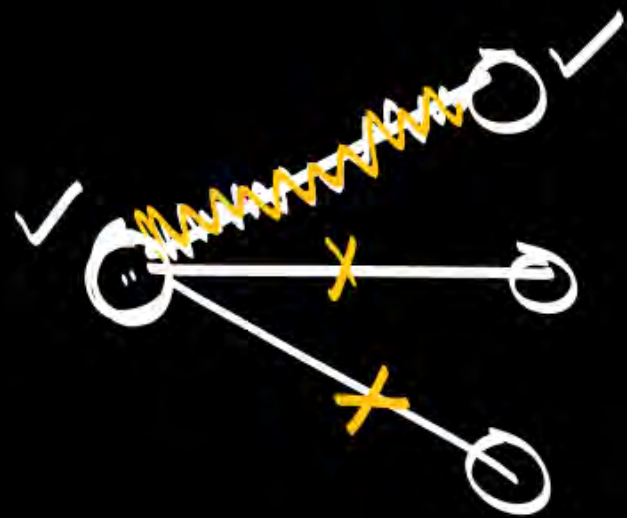
↓
collection of edges
where it should touch
all the vertices in the
graph.

maximal
matching set



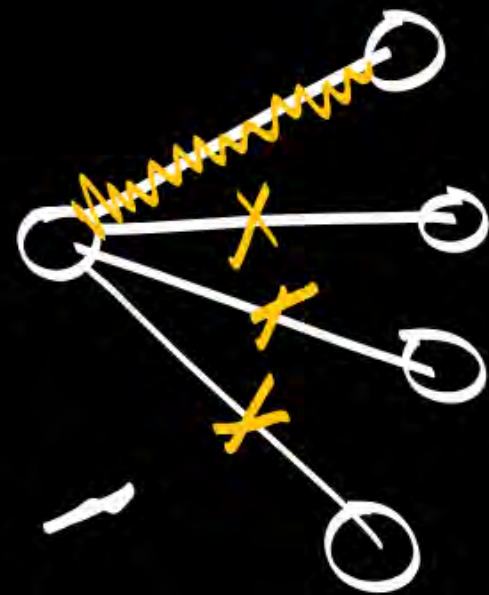
perfect matching:





{ maximal matching set

P.M X.



{ maximal matching set

P.M X.

Thm:

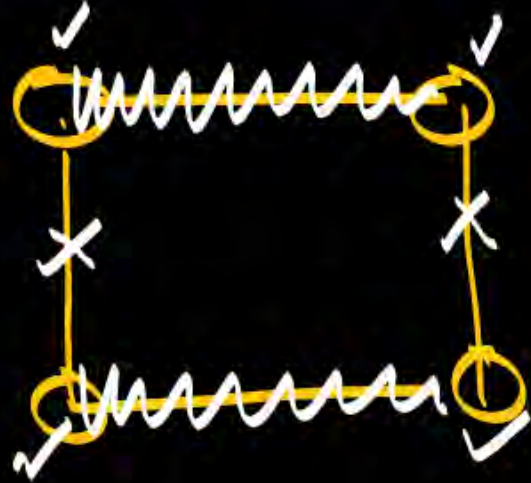
if perfect matching exist in graph, then no. of vertices will be even.

but vice versa is not true.

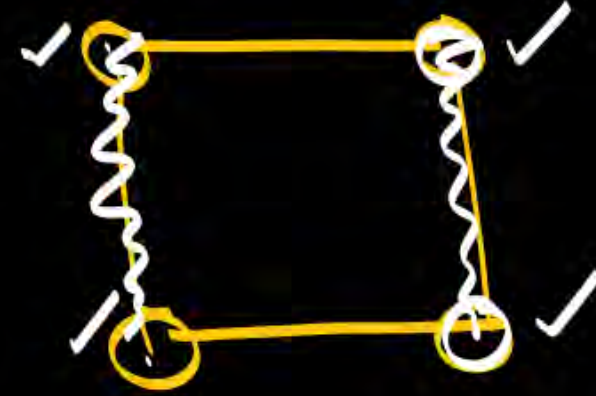
$n = 4$



P.M X.

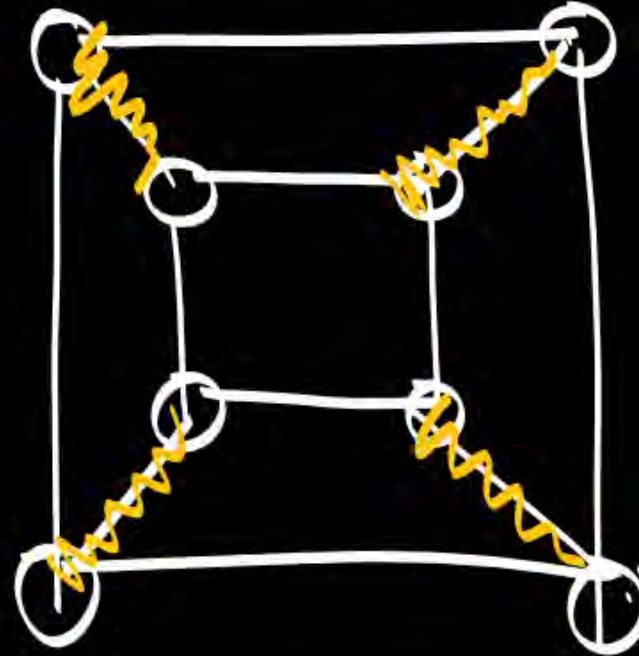
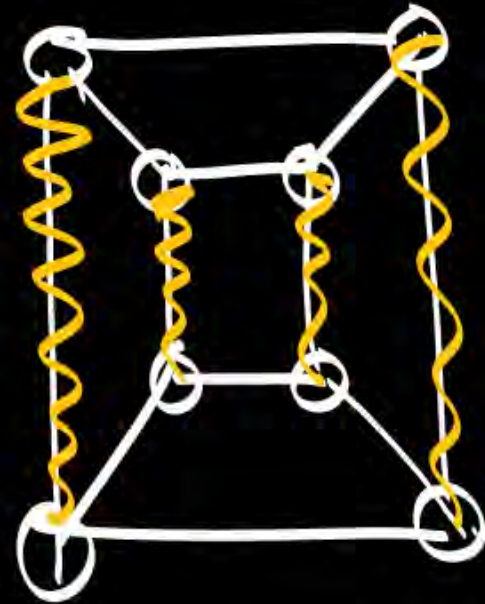
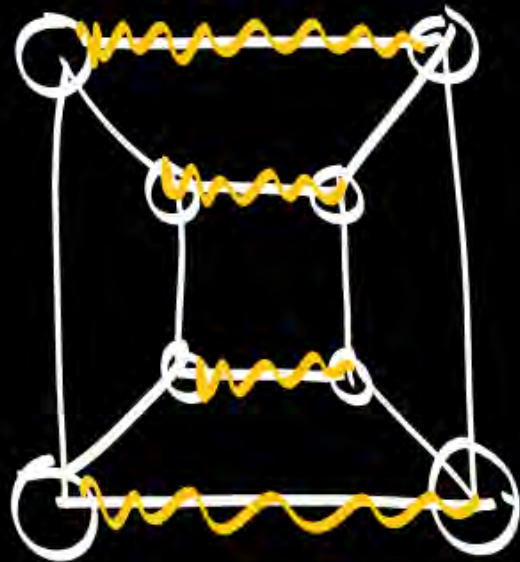


Type-1



Type-2

In Same graph we can have diff. types of perfect matching.



Complete Graph

K_{2n}

Total vertices = $2n$ (even)

Total vertices = $2n$.

$$(2n-1) \times (2n-3) \times (2n-5) \dots$$



$$(2n-1) \times (2n-3) \times (2n-5) \dots$$

$$\frac{2n}{2n} \times (2n-1) \times \frac{(2n-2)}{(2n-2)} \times (2n-3) \times \frac{(2n-4)}{(2n-4)} \times (2n-5)$$

$$(2n)(2n-1)(2n-2)(2n-3)(2n-4)(2n-5) = \underline{(2n)!}$$

$$2n \times (2n-2) \times (2n-4) \dots$$

take 2
common in DN

$$\frac{(2n)!}{2 \cdot n \cdot 2 \cdot (n-1) \dots} =$$

$$\frac{(2n)!}{2^n \cdot n!}$$

GATE

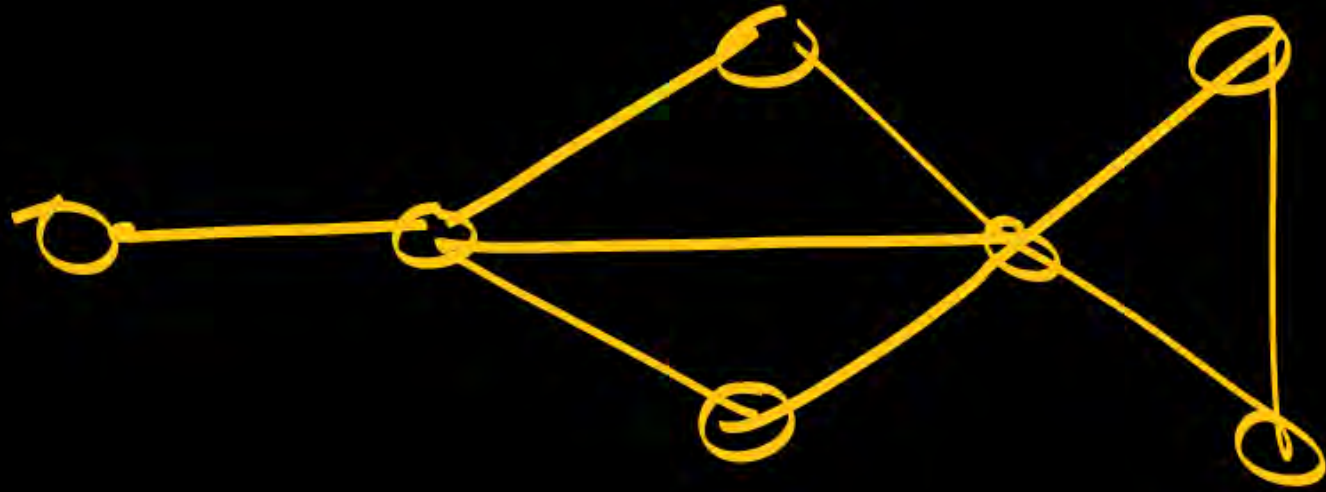
what will be total perfect matching in complete Graph of 6 vertices?

$$2n = 6$$

$$n = 3$$

$$\frac{(2n)!}{2^n \cdot n!} = \frac{(2 \cdot 3)!}{2^3 \cdot 3!} = \frac{6!}{8 \cdot 3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3!}{8 \cdot 3!} = \frac{6 \cdot 5 \cdot 4}{8} = 15$$

mistake.
 $n = 6$



Independence set: set of non adjacent vertices

maximal independence set:

Independence no ($\beta(G)$)

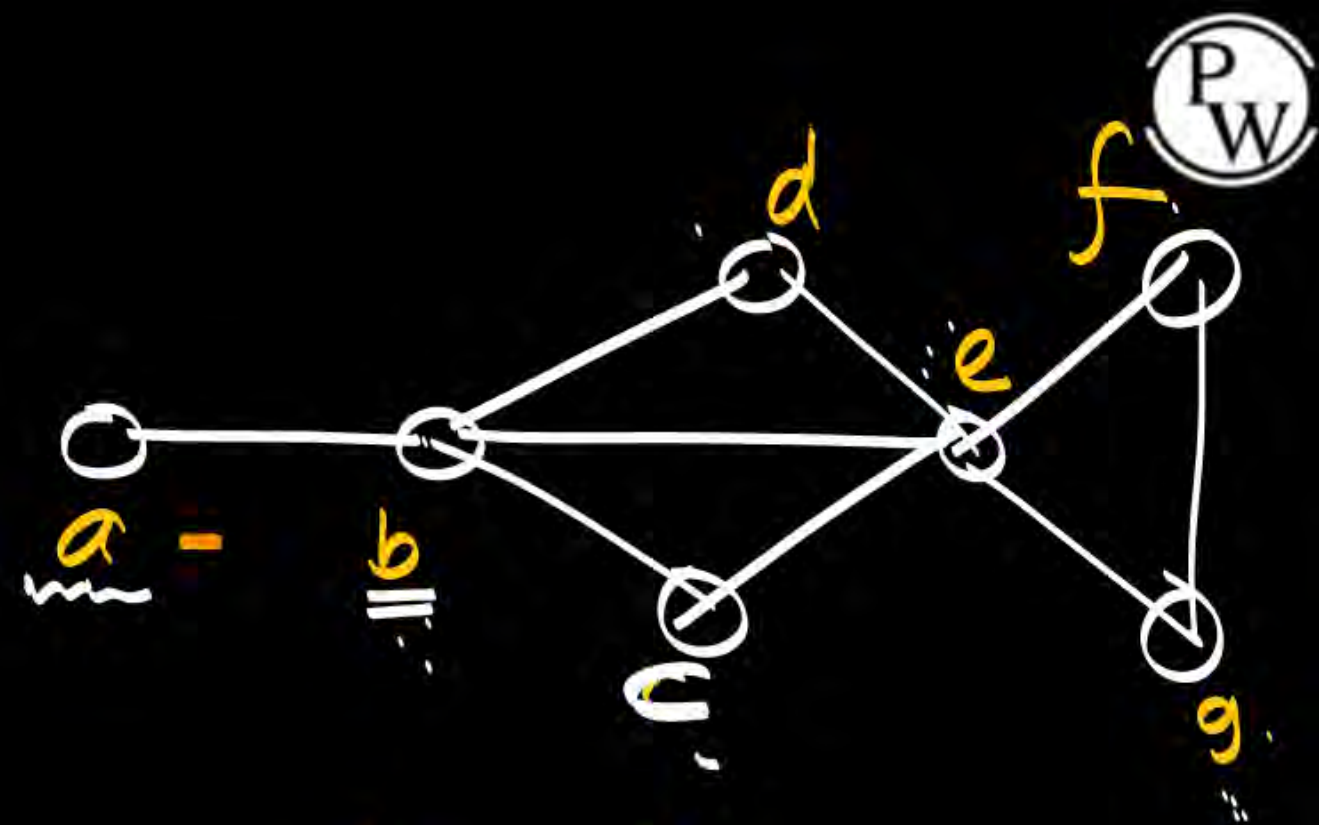
Dominance set :

Graph $G = (V, E)$

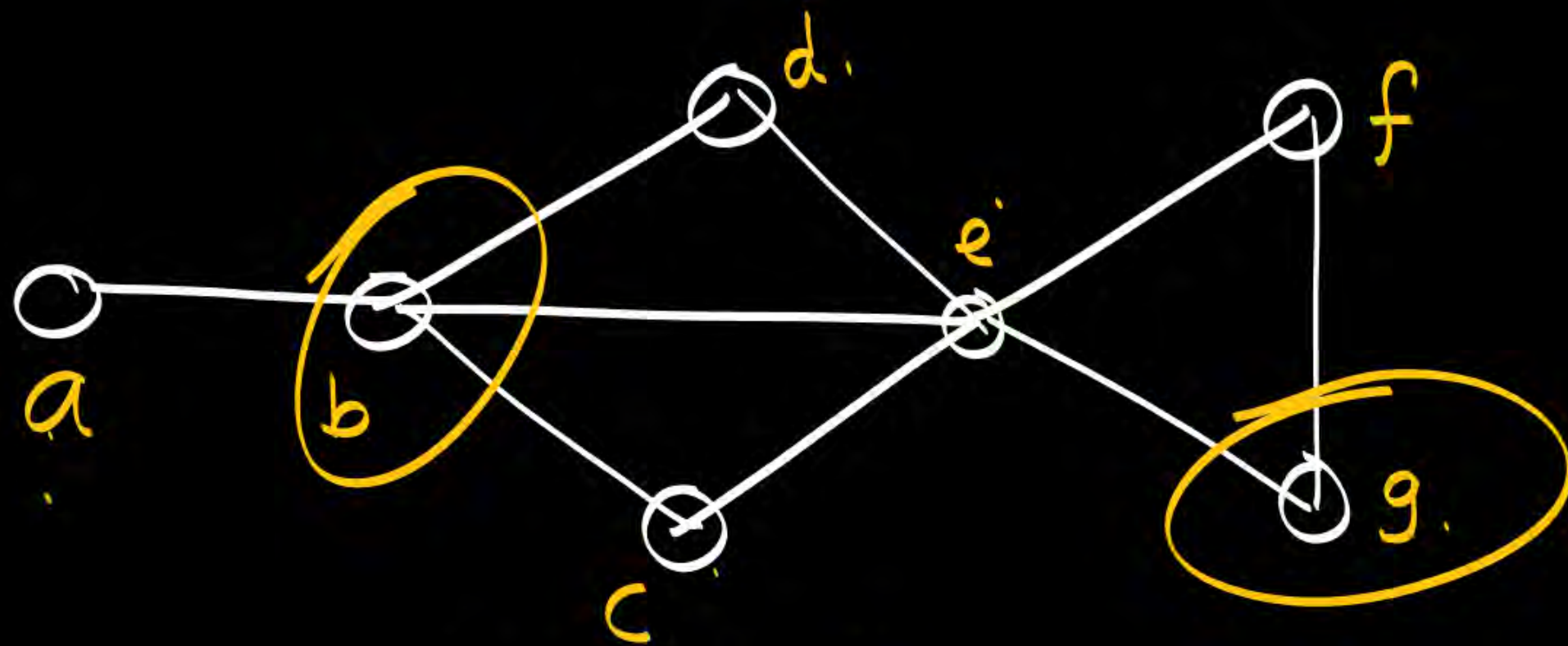
Dominance set D

$$D \subseteq V$$

if we take any vertex from V
either that vertex belongs to D
or it's adjacent belongs to D .



$D: \{a, b, e\}$



Dominating set
Such that we cannot
remove any new
vertex from this.

$D = \{a, b, c, d, e, f, g\}$ D_s

$\{b, c, d, e, f, g\}$

$\{b, c, d, e, f, g\} \rightarrow \{b, c, f, g\} \rightarrow \{b, f, g\} \rightarrow \{b, g\}$

minimal
dominating
set

In a Graph,
we get minimal dominating set
of diff sizes

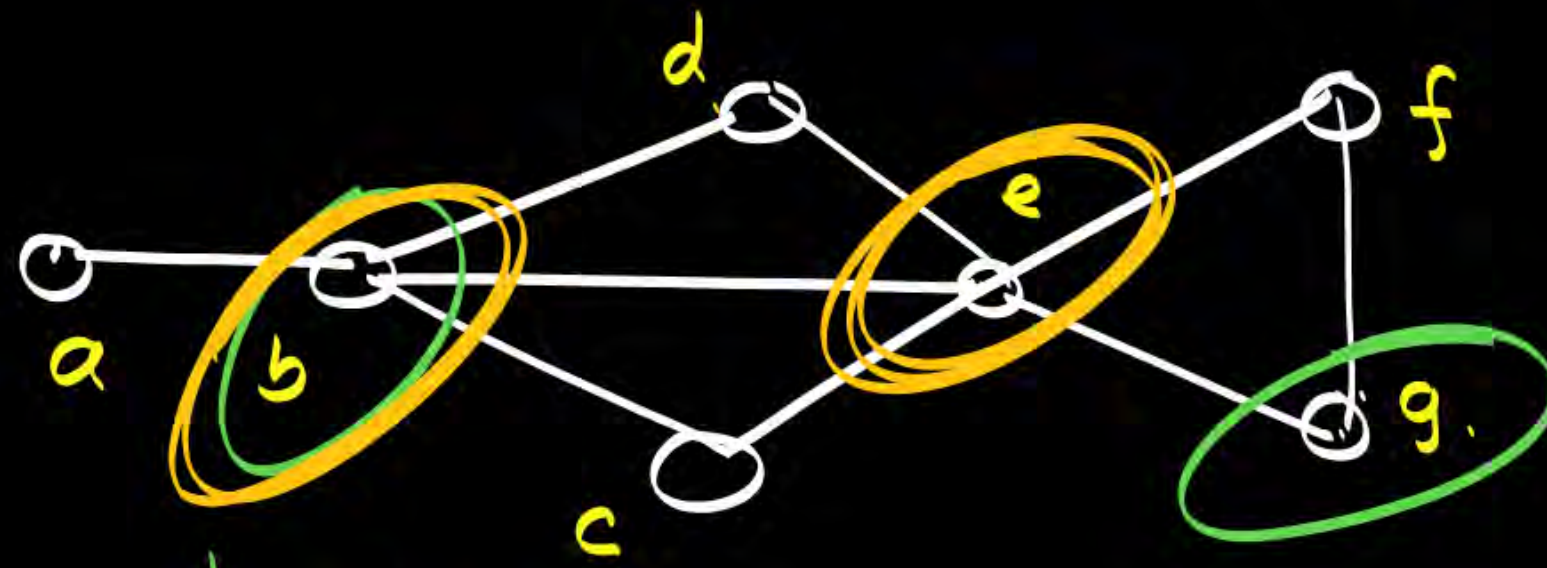
Dominance no ($\alpha(G)$)

no. of vertices present in
smallest minimal dominating set

m.d.s.

smallest { a, c, d, f }
 { a, c, d, g }
 { d, e } ✓
 { b, f } ✓
 { b, g } ✓

$\alpha(G) = 2$



$\{b, g\}$
MIS

MIS

$\{b, g\}$
 $\{b, f\}$
 $\{a, c, d, f\}$
 $\{a, e\}$

MDS

$\{b, g\}$
 $\{b, f\}$
 $\{a, c, d, f\}$
 $\{a, e\}$

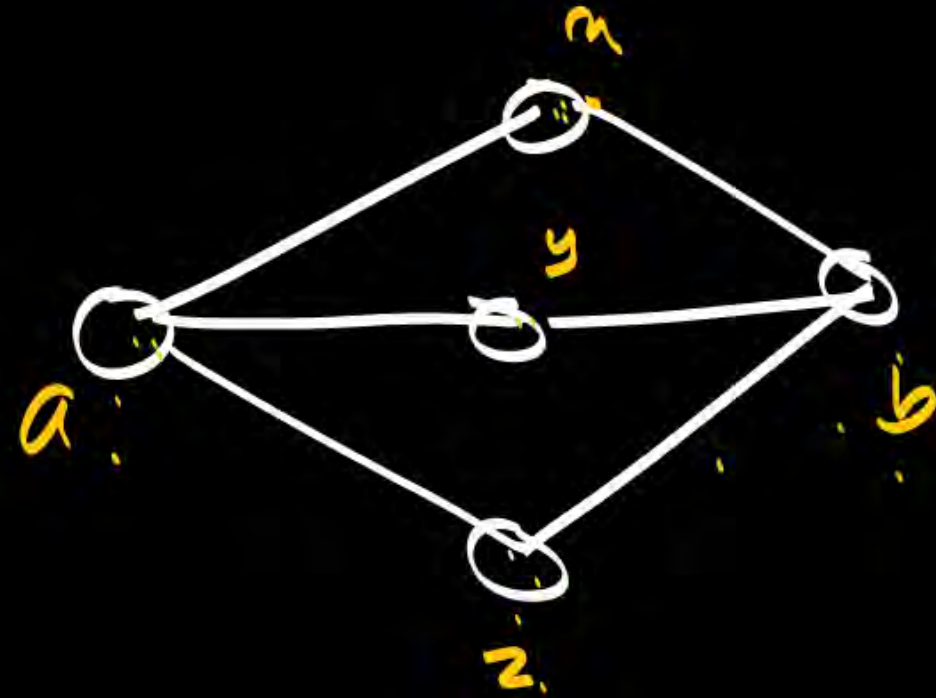
$\{b, e\}$

smallest Every maximal Independence set
will also be minimal dominating
set

but viceversa is not True.
eg: $\{b, e\}$

$$\alpha(G) \leq \beta(G)$$

max
 $\beta(G) = 4$

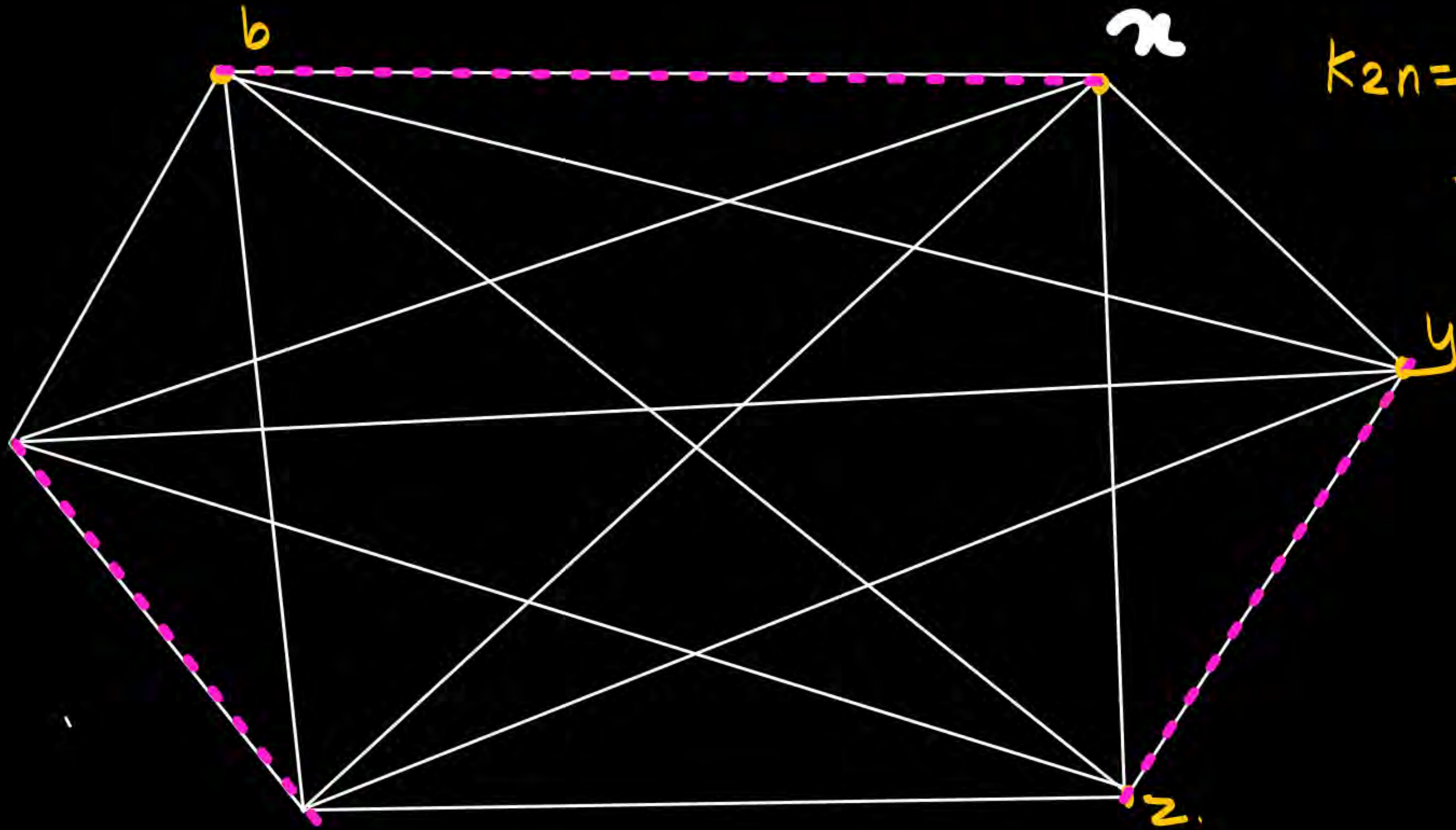


Dominating set

minimal Dominating set

$$D = \{a, b\}$$

ask 1, 2, 3.



$$K_{2n} = K_G$$

Total diff.
types of
perfect
matching.



2 mins Summary

Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five

Matching Set Part-02



THANK - YOU